# Ashlesha Ahirwadi

Chicago, IL | ahirwadiashlesha@gmail.com | +1 773 828 7560 | linkedin.com/in/ashlesha-ahirwadi | Portfolio

## Education

**Northwestern University**, Evanston, IL | *M.S. in Artificial Intelligence* Sept 2024 – Dec 2025

**Cummins College of Engineering for Women**, Pune, India | *B.Tech in Information Technology* Sept 2020 – Jun 2024

## Experience

**AI/ML Engineering**, Imagit, Inc. | Chicago, IL Aug 2025 – Present

- Built Pronto, Imagit's disaster recovery platform, designing an AI-powered quote recommendation system that takes client requirements as input and generates tailored recovery plan quotes, integrating sales, marketing, and finance data into a centralised CRM to give teams a unified view of client and operational data.
- Developed a multi-agent RAG system on top of the Pronto data layer, enabling teams to query and retrieve contextual information across CRM, sales, marketing, and finance platforms through natural language, reducing manual cross-platform lookups and accelerating decision-making for client-facing and internal workflows.
- Built an internal technician recruitment agent that autonomously sources candidates, sends screening links, evaluates responses, and manages onboarding, replacing a manual hiring workflow with an end-to-end agentic pipeline that operates with minimal human intervention.

**AI Engineering Intern**, Arqaios Inc. | Chicago, IL Jun 2025 – Aug 2025

- Built an MVP for a smart electric fixture using mmWave radar sensors and AI to enable real-time fall detection and human presence awareness, engineering motion and activity detection algorithms that process sensor features including range, azimuth, velocity, and confidence values to classify human activity states for elderly care safety applications.
- Designed a hybrid edge-cloud inference pipeline where the AI model runs on-device for responsive, low-latency alerts while streaming aggregated data to the cloud for diagnostics and monitoring, optimising fall detection models for elderly care use cases and ensuring scalability across smart home deployment environments.
- Collaborated with hardware and software teams to align mmWave signal acquisition standards with production deployment requirements, applying real-time ML inference techniques and contributing to the full system lifecycle from sensor data collection through model training, deployment, and iterative performance improvement.

## Projects

**GenAI Enterprise Solution: Azure ML, RAG Architecture and Agentic AI Orchestration**

- Developed a GenAI solution using Azure Machine Learning and RAG architecture to automate enterprise operational workflows, implementing agentic frameworks where AI agents autonomously plan and execute multi-step tasks with prompt engineering pipelines to optimise LLM output quality across diverse business use cases.
- Deployed scalable cloud infrastructure on Azure (AKS, CI/CD pipelines), applying GenAI Ops and enterprise-grade AI orchestration practices to ensure the platform remains reliable, observable, and maintainable at scale from proof of concept through production.

**MediCurious: Specialised Medical LLM with Scalable Data Pipeline and Live Agentic Deployment**

- Developed a Large Language Model specialised for healthcare decision support, trained to provide treatment recommendations, symptom analysis, and medication safety evaluations by designing a scalable data pipeline integrating heterogeneous medical datasets into optimised JSONL formats for efficient LLM fine-tuning.
- Deployed a live real-time agentic system for AI-augmented clinical decision support, demonstrating the end-to-end journey from raw data ingestion through model training and production deployment in a regulated domain.

**Stock Market Prediction: Options Trading Forecasting on NASDAQ 100 (2009–2024)**

- Developed predictive ML models for options trading on 15 years of NASDAQ 100 time series data, benchmarking SARIMAX, LSTM, TiDE, and TS-Mixer architectures to forecast stock price trends and identify optimal trading strategies, resolving volume misrepresentations to improve data quality and model accuracy.
- Delivered structured performance visualisations and comparative reports across model architectures, providing insights into predictive performance across different market conditions to support data-driven trading decisions.

**Lung Tumor Detection from 3D CT Scans: Northwestern Medicine, Abazeed Lab**

- Built a production-ready deep learning ensemble (YOLO3D, UNETR, VNet) for radiation therapy planning at Northwestern Medicine, achieving 75.6% sensitivity and 69.6% AP@0.3 with custom CIoU loss and anisotropic anchor designs tailored for medical precision.
- Accelerated model training on H100 GPUs using mixed-precision, 16+ DataLoader workers, PyTorch Lightning, and MONAI medical imaging transforms, making rapid clinical iteration feasible at production scale.

## Technologies

**Languages:** Python, SQL, Scala, Java, JavaScript, HTML/CSS, C **AI Coding:** Claude Code, Cursor

**AI/ML & GenAI:** PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace, LangChain, LangGraph, AutoGen, LlamaIndex, RAG Pipelines, Multi-Agent Systems, LLM Fine-tuning, Agentic AI, Prompt Engineering, MONAI, NLTK, spaCy, Pandas, NumPy

**Cloud & MLOps:** Azure ML, Azure AKS, AWS (Bedrock, EMR), CI/CD (GitHub Actions, Azure DevOps), Docker, Kubernetes, MLflow, Vector Databases (Chroma, Pinecone), Git